INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

IGNITION ELEMENT, ELECTRIC, M55 ASSEMBLY

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M55 Ignition Element Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 Quality Assurance Terms and Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1377

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-1168	Ammunition Lot Numbering

(Unless otherwise specified, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head Division (CAGE Codes 19200 and 19203)

DL10534225	Ignition Element, Electric, M55 Assembly
8887525	Box, Packing, Ammunition, for Ignition Element, Electric, M55
9233342	Container, Ammunition, Metal, for Ignition Element, Electric, M55

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ANSI B46.1	Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

3.2 Conformance to documents. The M55 Ignition Element covered by this specification shall be manufactured in accordance with DL10534225.

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3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Electrical.

3.4.1 Dielectric withstanding voltage. The body assembly, prior to the assembly of the bridgewire, shall not exhibit a leakage current in excess of 100 microamperes when subjected to 1000 volts of direct current applied between pin and body.

3.4.2 Bridge circuit resistance. The bridge circuit resistance of the ignition element shall be 0.11 ± 0.03 ohm.

3.5 Environmental.

3.5.1 Leakage. The ignition element shall not exhibit a leak rate of 1×10^{-4} cubic centimeters per second or more when withstanding a dry leak test at a pressure differential of 14 psi.

3.5.2 Vibration. The ignition element shall withstand vibration testing in accordance with MIL-STD-810, method 514.3, test procedure I, test condition I, with no evidence of disintegration of the propellant or derangement of components. The ignition element shall not autoignite, shall be safe to handle, and shall meet the requirements of 3.4.2, 3.5.1, 3.7.1, and 3.7.2 at 70°F following vibration testing.

3.5.3 Hydrostatic pressure. The body assembly shall be capable of withstanding a hydrostatic pressure of 10,000 psi without any evidence of mechanical failure or leakage (see 6.4.2).

3.6 Radiographic examination. Each ignition element shall be radiographically examined for defects (see 4.5.3.4).

3.7 Functional.

3.7.1 Maximum no-fire stimulus. The electrical ignition element shall not fire when a direct current of 1.5 amperes, maximum, is applied for 60 seconds.

3.7.2 All-fire stimulus. After the application of the maximum no-fire stimulus current, the electrical ignition element shall fire within 50 milliseconds maximum when a direct current of 5.0 amperes is applied at temperatures of −65°F, 70°F, and 200°F. The test fixture shall be temperature conditioned along with the test unit. The electric ignition element shall deliver a pressure of 2000 psi minimum to 6000 psi maximum when fired into a 1 cubic centimeter bomb. There shall be no evidence of blowback or mechanical failure of the glass-to-metal seal (see 6.4.1 and 6.4.2).

3.7.3 Minimum all-fire stimulus. The ignition element shall fire within 50 milliseconds when a direct current of 3.5 amperes is applied at temperatures of −65°F, 70°F, and 200°F. The electric ignition element shall meet the requirements of 3.7.2. Data will be recorded for information only (see 6.4.1).

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3.8 Workmanship. Workmanship shall be in accordance with the requirements of the applicable drawing, reference specifications, and the following.

3.8.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.8.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550 and MIL-STD-453.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified in the contract or purchase order (see 6.2), inspection conditions shall be as follows:

a. Temperature: Room ambient 68 to 77°F
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric: 14.7 psia

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4.4 First article inspection. At the beginning of regular production, the contractor shall submit a first article sample to the Government approved facility for evaluation. The sample shall consist of 112 ignition elements, 10 sets of parts, and 5 inert body assemblies. The sample shall be subjected to the tests in 4.5.3. Test procedures shall be in accordance with 4.6. The first article sample shall be manufactured using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system. The Government reserves the right to require new first article samples until such time as an acceptable sample is submitted (see 6.5).

4.4.1 Examination, tests, and facility. After provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or other such facility specified in the contract (see 6.2).

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.5.1.2 Lot size. The production lot size shall be as specified in the contract (see 6.2). The maximum lot size allowed is 2400 exclusive of test samples and shall consist of:

a. Components produced by a homogeneous process

b. Body assemblies (metal-to-ceramic seal) which have been heated in a furnace with the same process control

c. Primer mixtures from not more than one lot

d. Loading produced by a homogeneous process

4.5.2 Examination. One hundred percent examination shall be performed for all non-destructive critical defects. Examination for all major and minor defects shall be performed in accordance with the classification of defects. Four power (4x) magnification may be used for inspection of such defects as burrs and foreign matter. When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All non-conforming material shall be rejected.

4.5.3 Inspection testing. The following tests shall be performed in accordance with the provisions of 4.6.

4.5.3.1 Electrical.

4.5.3.1.1 Dielectric withstanding voltage. Each body assembly prior to assembly of bridgewire shall be subjected to the test. Failure of any unit to comply with the requirements of 3.4.1 shall be cause for rejection of the unit.

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4.5.3.1.2 Bridge circuit resistance. Each ignition element assembly shall be subjected to the test. Failure of any unit to comply with the requirements of 3.4.2 shall be cause for rejection of the unit.

4.5.3.2 Environmental.

4.5.3.2.1 Leakage. Each ignition element shall be subjected to the test. Failure of any unit to comply with the requirements of 3.5.1 shall be cause for rejection of the unit.

4.5.3.2.2 Vibration. Twelve ignition assemblies of the initial production sample shall be subjected to the test. Failure of any unit to comply with the requirements of 3.5.2 shall be cause for rejection of the lot. Any changes in the loading process shall be cause for selecting a new sample for the vibration test.

4.5.3.3 Hydrostatic pressure. Five body assemblies of each lot shall be subjected to the test. Failure of any unit to comply with the requirements of 3.5.3 shall be cause for rejection of the lot.

4.5.3.4 Radiographic examination. All assembled ignition elements shall be radiographically examined in accordance with MIL-STD-453, with the exception that an inert, unplated ignition element shall be used as the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection of the item. Ignition elements shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. Ignition elements shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive order beginning with the number 001 in each production lot. Any missing numbers are to be identified on the X-rays. The ignition elements shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic plate identification shall include item nomenclature, the complete lot number, the span of serial numbers displayed, and the contract number. The radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective ignition elements found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of all parts and explosives.

4.5.3.5 Functional. The following tests (4.5.3.5.1 through 4.5.3.5.3) shall be performed by the Government. One hundred ignition elements shall be selected from each lot, assembled into a 1-cubic- centimeter firing chamber and temperature conditioned in accordance with table I and 4.6.4.4.

TABLE I. Temperature conditioning.

Temperature (°F)	Number of units
-60 to -70	20
65 to 75	60
195 to 205	20

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4.5.3.5.1 Maximum no-fire stimulus. Fifty ignition elements (30 at 65 to 75°F, 10 at –65 to –70°F, and 10 at 195 to 200°F) from each lot shall be tested. Failure of any unit to comply with the requirements of 3.7.1 shall be cause for rejection of the lot.

4.5.3.5.2 All-fire stimulus. The 50 ignition elements tested in 4.5.3.5.1 shall be used for this test. Failure of any unit to comply with 3.7.2 shall be cause for rejection of the lot.

4.5.3.5.3 Minimum all-fire stimulus. The remaining 50 conditioned ignition elements (30 at 65 to 75°F, 10 at –65 to –70°F, and 10 at 195 to 200°F) shall be used for this test.

4.6 Test methods and procedures. All tests shall be performed on units which have been visually and dimensionally accepted. There shall be no dirt, oils, foreign matter, defective materials. Markings shall be clear, legible, and correct.

4.6.1 Electrical.

4.6.1.1 Dielectric withstanding voltage. A dielectric withstanding voltage of 100 + 50 volts dc shall be applied between the pin and body prior to the assembly of the bridgewire. In each test, the leakage current shall be measured within an accuracy of 5 percent. The test shall be conducted in accordance with MIL-STD-202, Method 301.

4.6.1.2 Bridge circuit resistance. The bridge circuit resistance of the bridge circuit shall be measured using a test circuit which subjects the bridge circuit to a current of less than 50 milliamperes. The test shall be conducted in accordance with MIL-STD-202, Method 303.

4.6.2 Environmental.

4.6.2.1 Leakage. The leak test shall be conducted using a leak detector of sufficient sensitivity to ascertain if the ignition elements meet the leak rate requirement.

4.6.2.2 Vibration. The ignition elements shall be subjected to the test requirements of 3.5.2. The sample shall be divided into three groups of four units each. The first group shall be tested with disc up, the second group with disc down, and the third group with disc horizontal. If the ignition elements have successfully met the requirements of the vibration test, they shall then be subjected to tests as outlined in 4.6.1.2 and 4.6.2.1. The ignition elements shall be assembled into a 1-cubic-centimeter firing chamber and temperature conditioned at 65 to 75°F in accordance with 4.6.4.4. After temperature conditioning, the ignition elements shall be subjected to tests in the following order: (1) 4.6.4.1 and (2) 4.6.4.2.

4.6.3 Hydrostatic pressure. The test shall be conducted using a hydrostatic test apparatus capable of subjecting the body assembly to an internal pressure of 10,000 psi for a minimum of 10 seconds.

4.6.4 Functional.

4.6.4.1 Maximum no-fire stimulus. A dc current pulse of 1.50 – 0.05 amperes shall be applied to the bridge circuit of each electric ignition element for 60 ± 2 seconds.

4.6.4.2 All-fire stimulus. Within 4 ± 1 minutes after successfully completing the test in 4.6.4.1, a dc current pulse of 5.0 + 0.05 amperes shall be applied to the bridge circuit of the ignition element. The time from

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current application until the ignition element fires and the maximum pressure attained within the firing chamber shall be measured and recorded. Temperature reconditioning is required when more than 5 minutes lapse after the maximum no-fire stimulus test.

4.6.4.3 Minimum all-fire stimulus. A dc current pulse of 3.50 + 0.05 amperes shall be applied to the bridge circuit of each ignition element. The time from current application until the ignition element fires and the maximum pressure attained within the firing chamber shall be measured and recorded.

4.6.4.4 Temperature conditions. The conditioning chamber shall be preconditioned to the specified temperature. Once the temperature inside the chamber has stabilized, the samples shall be placed inside and maintained at the specified temperature for 4 hours minimum, 24 hours maximum.

4.7 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or as otherwise specified by the procuring activity (see 6.2). Packing and marking shall be in accordance with Drawings 9233342, 8887525, and MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The ignition element is used as a secondary emergency release in bomb racks and launchers to free stores in an emergency.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, date, and revision letter of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced and that MIL-STD-810D shall be used (see 2.1, 3.5.2 and 4.7.4).

c. Whether a first article sample is required (see 3.1, 4.4 and 6.5).

d. Test facility for first article tests (see 4.4.1).

e. Production test activity and lot size (see 4.1 and 4.5.1.2).

f. Any special test conditions (see 4.3).

g. Directions for shipping radiographs of entire lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.5.3.4).

h. Level of packing and marking.

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j. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" OP 5 are applicable.

6.3 Hazard notice. The ignition element described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.4 Definitions.

6.4.1 Ignition delay time. Ignition delay time is the time from first application of current to the start of pressure rise.

6.4.2 Mechanical failure. Mechanical failure is any deformation or breakage of a part other than a design failure mode.

6.5 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Subject term (keyword) listing.

Stores release
Bomb rack
Rocket launcher
Electric detonation

6.7 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Preparing activity:
Navy - OS
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